
Conditioning, covariance, and transformations

Antonio de Leon

Probability and Statistics for Engineers Fall 2026

Monday, November 2

Conditional densities and means

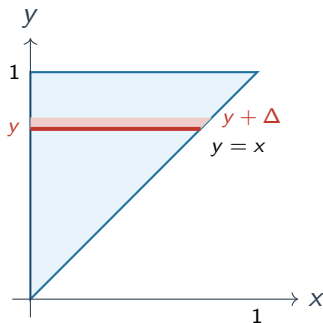
Wednesday, November 4

Covariance and variance of sums

Friday, November 6

Transformations using CDFs

A fixed coordinate



Last week:

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$f_X(x) = 2(1 - x), \quad 0 < x < 1,$$

$$f_Y(y) = 2y, \quad 0 < y < 1.$$

Each marginal is zero elsewhere.

If the later time is y , where can the earlier time lie?

Normalize a density slice

Fix y with $0 < f_Y(y) < \infty$. Seek a density in x of the form

$$f_{X|Y}(x | y) = c(y)f_{X,Y}(x, y).$$

Choose $c(y)$ so the slice has total probability one.

This defines a conditional density. It does not divide by $P(Y = y)$, which is zero.
For continuous slices, conditioning on a narrow strip around y leads to the same ratio.

A conditional density is a density

For $0 < f_Y(y) < \infty$:

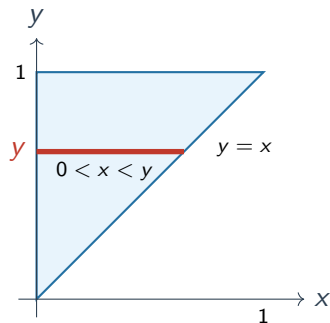
$$f_{X|Y}(x | y) \geq 0, \quad \int_{\mathbb{R}} f_{X|Y}(x | y) dx = 1.$$

$$P(X \in A | Y = y) = \int_A f_{X|Y}(x | y) dx.$$

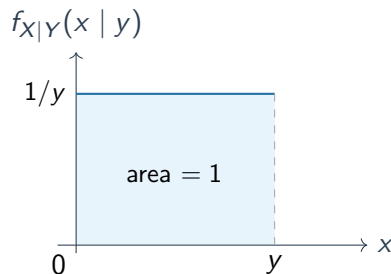
Multiplying by the marginal restores the joint density.

Equalities of densities are understood almost everywhere; zero-marginal values need no ratio.

Earlier time, given the later time



$$f_{X|Y}(x | y) = \begin{cases} \frac{1}{y}, & 0 < x < y, \\ 0, & \text{otherwise,} \end{cases} \quad 0 < y < 1.$$



The joint height is 2; normalization changes the height to $1/y$. Thus $X | Y = y$ is uniform on $(0, y)$.

A probability inside the slice

The later time is $Y = 3/4$. Find $P(X < 1/2 \mid Y = 3/4)$. The conditional density is $4/3$ on $(0, 3/4)$, zero elsewhere.



Compare with the marginal probability:

Fixing the later time changes the law of the earlier time.

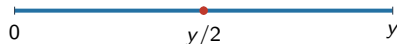
Mean after fixing $Y = y$

The conditional mean is a function of the fixed value:

$$m(y) = \mathbb{E}[X \mid Y = y] = \int_{\mathbb{R}} x f_{X|Y}(x \mid y) dx.$$

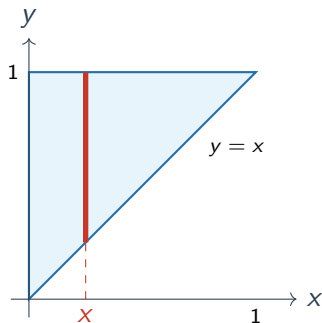
Require $\int |x| f_{X|Y}(x \mid y) dx < \infty$.

For the triangle, derive $m(y)$ and evaluate $m(3/4)$.



$m(y)$ is a number for fixed y . The notation $\mathbb{E}[X \mid Y] = m(Y) = Y/2$ denotes a random variable.

Reverse the conditioning



Fix $0 < x < 1$. Determine:

- ① the allowed values of y ;
- ② $f_{Y|X}(y | x)$;
- ③ $\mathbb{E}[Y | X = x]$.

Use $f_X(x) = 2(1 - x)$ in the denominator.

Work for three minutes; write the interval before the formula.

Check the reverse slice

$$\mathbb{E}[Y \mid X = x] = \int_x^1 \frac{y}{1-x} dy = \frac{1-x^2}{2(1-x)} = \frac{1+x}{2}, \quad 0 < x < 1.$$

Fixed coordinate	Remaining interval	Density	Mean
$Y = y$	$0 < x < y$	$1/y$	$y/2$
$X = x$	$x < y < 1$	$1/(1-x)$	$(1+x)/2$

Reversing the condition changes both the interval and the normalizing factor.

Conditioning under independence

A separate model: $f_{R,S}(r,s) = r$ on $0 < r < 1$, $0 < s < 2$, zero elsewhere.

$$f_R(r) = 2r \ (0 < r < 1), \quad f_S(s) = \frac{1}{2} \ (0 < s < 2).$$

The support is a rectangle, and the density factors there and off it.

Under independence, knowing $S = s$ leaves the distribution of R unchanged.

Choose the law before integrating

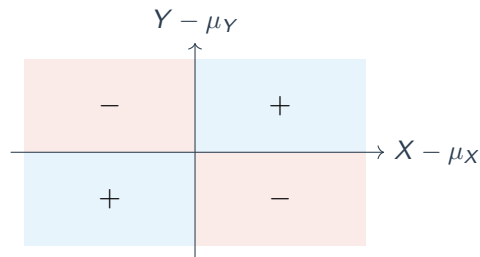
Question	Use this density	Integrate over
$P(X \in A)$	$f_X(x)$	A
$P(X \in A \mid Y = y)$	$f_{X,Y}(x, y)/f_Y(y)$	A within the slice
$\mathbb{E}[X \mid Y = y]$	$f_{X Y}(x \mid y)$	the full x -slice

In the triangle, suppose $0 < y < 1/2$. What is $P(X < 1/2 \mid Y = y)$?

Why can $\mathbb{E}[X \mid Y = 3/4]$ differ from $\mathbb{E}[X]$?

Products of deviations

Write $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Assume finite second moments.



$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

Same-sign deviations contribute positively.
Opposite-sign deviations contribute negatively.

Covariance averages these signed products;
cancellation is possible.

From the definition to a usable formula

Expand the product, then use linearity of expectation:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X), \quad \text{Cov}(X, X) = \text{Var}(X).$$

Units multiply: if both variables measure seconds, covariance is in seconds squared.

Covariance in the triangle

	First moment	Second moment
X	$\int_0^1 x 2(1-x) dx = 1/3$	$\int_0^1 x^2 2(1-x) dx = 1/6$
Y	$\int_0^1 y 2y dy = 2/3$	$\int_0^1 y^2 2y dy = 1/2$

A small cell contributes xy times its probability, approximately $xyf_{X,Y}(x,y)\Delta x\Delta y$. Summing gives the double integral.

Variance of a linear combination

Let $L = aX + bY + c$. Start by subtracting its mean:

$$L - \mathbb{E}[L] = a(X - \mu_X) + b(Y - \mu_Y).$$

Square before taking expectation. Keep the cross term.

No independence assumption was used.

Variation in elapsed time

In the triangle, $D = Y - X$ is the elapsed time between two ordered times.

$$\text{Var}(X) = \text{Var}(Y) = \frac{1}{18}, \quad \text{Cov}(X, Y) = \frac{1}{36}.$$

Find $\text{Var}(Y - X)$ and compare it with $\text{Var}(X + Y)$.

Positive covariance reduces the variance of a difference and increases the variance of a sum.

Why independence removes the term

For independent variables with a joint density,

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) \quad \text{almost everywhere.}$$

For discrete variables, the same argument uses double sums.

Finite second moments ensure $\mathbb{E}[|XY|] < \infty$, so this factorization is justified.

Zero covariance can coexist with dependence

Let A be uniform on $\{-1, 0, 1\}$ and set $B = A^2$.

A	-1	0	1
B	1	0	1
AB	-1	0	1
Probability	1/3	1/3	1/3

The signed products cancel, even though B is determined by A .

Covariance on a common scale

For $0 < \sigma_X^2 = \text{Var}(X) < \infty$ and $0 < \sigma_Y^2 = \text{Var}(Y) < \infty$,

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \mathbb{E}[Z_X Z_Y], \quad Z_X = \frac{X - \mu_X}{\sigma_X}, \quad Z_Y = \frac{Y - \mu_Y}{\sigma_Y}.$$

Correlation is unitless. If either variance is zero, this ratio is undefined.

More than two variables

Write $W_i = X_i - \mathbb{E}[X_i]$. Each pair appears twice in the square:

$$\left(\sum_{i=1}^n W_i \right)^2 = \sum_{i=1}^n W_i^2 + 2 \sum_{i < j} W_i W_j.$$

Pairwise zero covariance is sufficient for variances to add. Independence is a stronger sufficient condition.

For three or more variables, covariance terms can also cancel in the total; each term need not be zero.

Averaging repeated measurements

Suppose $X_1, \dots, X_n$ are independent, each with variance σ^2 .

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

What if every recorded value is the same random measurement, $X_i = X_1$?

Additional measurements reduce variance only when their dependence permits it.

Keep the covariance term

For the triangle, find $\text{Var}(2X - Y + 3)$.

$$\text{Var}(X) = \text{Var}(Y) = \frac{1}{18}, \quad \text{Cov}(X, Y) = \frac{1}{36}.$$

Square the individual coefficients; preserve their signs in the cross term. The constant does not affect variance.

Start with the transformed event

For $T = g(X)$, begin with

$$F_T(t) = P(T \leq t) = P(g(X) \leq t).$$

- 1 Find the possible values of T .
- 2 Solve $g(x) \leq t$ for the original variable x .
- 3 Compute the probability under the law of X .
- 4 Write the full CDF, including values outside the range.
- 5 If the law is absolutely continuous, differentiate for a density.

Last week's normal standardization also worked by rewriting an event.

A CDF describes every distribution. Its derivative alone may miss an atom.

Transform a marginal law

Return to the triangle: $f_Y(y) = 2y$ for $0 < y < 1$. Let $Z = Y^2$.

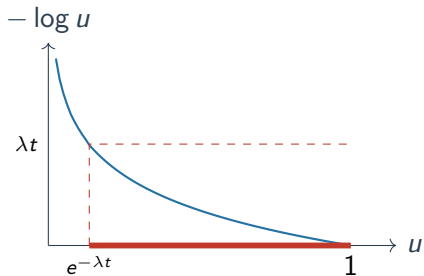
$$0 < Z < 1, \quad \{Y^2 \leq z\} = \{Y \leq \sqrt{z}\} \quad (0 < z < 1).$$

A function of Y alone needs the marginal law of Y .

A decreasing transformation

Let $U \sim \text{Uniform}(0, 1)$ and $\lambda > 0$. Set

$$T = -\frac{\log U}{\lambda}.$$



Multiplying by -1 reverses the inequality.

For $t \geq 0$, rewrite the event carefully.

Recover the exponential law

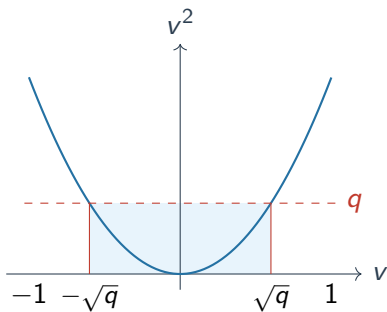
For $t \geq 0$, $F_T(t) = P(U \geq e^{-\lambda t})$.

If $\lambda = 2$, find $P(T > 1/2)$.

The uniform input produces the exponential waiting-time law from last week.

Squaring has two preimages

Let $V \sim \text{Uniform}(-1, 1)$ and $Q = V^2$.



For $0 < q < 1$, identify all original values satisfying $V^2 \leq q$.

Both the negative and positive branches contribute probability. The image is $[0, 1)$; $P(Q = 0) = 0$.

The law of the square

For $0 < q < 1$, integrate over both preimages:

$$F_Q(q) = \int_{-\sqrt{q}}^{\sqrt{q}} \frac{1}{2} dv.$$

An unbounded density near zero does not create a point mass: the CDF is continuous there.

Transform a square root

Let $U \sim \text{Uniform}(0, 1)$ and $W = \sqrt{U}$.

- ① State the possible values of W .
- ② Find the full CDF of W .
- ③ Find a density and check its integral.

Work for three minutes. Solve the event before differentiating.

Check the transformed law

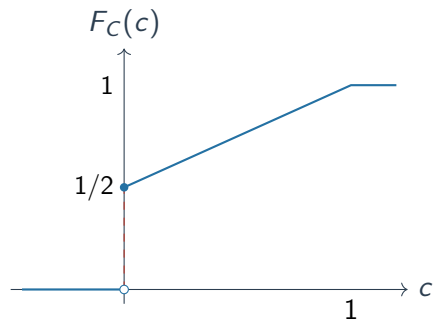
$$F_W(w) = \begin{cases} 0, & w \leq 0, \\ w^2, & 0 < w < 1, \\ 1, & w \geq 1, \end{cases} \quad f_W(w) = \begin{cases} 2w, & 0 < w < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Compute $\mathbb{E}[W]$ in two ways.

The two calculations must agree: both average the same random quantity.

A transformation can create an atom

Let $V \sim \text{Uniform}(-1, 1)$ and $C = \max(0, V)$.



$$F_C(c) = \begin{cases} 0, & c < 0, \\ (c + 1)/2, & 0 \leq c < 1, \\ 1, & c \geq 1. \end{cases}$$

A whole input interval collapses to zero. The jump records the missing probability.

Check the event and the endpoints

Transformation	Interior threshold	Original event
$Y^2, Y > 0$	$0 < z < 1$	$Y \leq \sqrt{z}$
$-\log U/\lambda$	$t > 0$	$U \geq e^{-\lambda t}$
$V^2, -1 < V < 1$	$0 < q < 1$	$-\sqrt{q} \leq V \leq \sqrt{q}$
$\max(0, V)$	$0 \leq c < 1$	$V \leq c$

For each CDF: check the range, every preimage, both tails, and any jumps.

Optional: average the conditional means

Assume $\mathbb{E}[|X|] < \infty$. Then $m(y) = \mathbb{E}[X \mid Y = y]$ is integrable under f_Y .

Each conditional mean is weighted by how often its conditioning values occur.

Optional: decompose the variance

Assume $\mathbb{E}[X^2] < \infty$. Define

$$m(Y) = \mathbb{E}[X \mid Y], \quad v(Y) = \text{Var}(X \mid Y).$$

Average variation within a slice, plus variation between the slice means.

Optional: variance within and between slices

In the triangle, $X \mid Y = y \sim \text{Uniform}(0, y)$.

$$m(y) = \frac{y}{2}, \quad v(y) = \frac{y^2}{12}.$$

Contribution	Calculation	Value
Within slices	$\mathbb{E}[Y^2/12] = (1/12)(1/2)$	1/24
Between slice means	$\text{Var}(Y/2) = (1/4)(1/18)$	1/72

Both terms are nonnegative and together recover the marginal variance.

Optional: a smooth monotone inverse

Let X have a density on an interval. Suppose $T = g(X)$, where g is continuously differentiable, strictly monotone, and $g'(x) \neq 0$. Write $h = g^{-1}$.

$$T = aX + b, \quad a \neq 0: \quad f_T(t) = \frac{1}{|a|} f_X\left(\frac{t - b}{a}\right).$$

Use the original CDF method for multiple branches or collapsed intervals.

Optional: binomial variance from indicators

Let $I_1, \dots, I_n$ be independent Bernoulli(p) variables and $N = \sum I_i$. Since $I_i^2 = I_i$,

$$\mathbb{E}[I_i] = p, \quad \text{Var}(I_i) = \mathbb{E}[I_i^2] - p^2 = p(1 - p).$$

A fixed success probability is essential to this calculation; a randomly varying probability gives a different model.

Reading selections

DeGroot and Schervish, *Probability and Statistics*, 4th ed.

- Conditioning: pp. 144–146; conditional means, pp. 256–257.
- Covariance and correlation: pp. 248–252; variance of sums, pp. 253–254.
- CDF transformations: pp. 168–169; atoms under transformations, p. 172.

Harchol-Balter, *Introduction to Probability for Computing*, 2024 text.

- Conditional density and mean: Definitions 8.9 and 8.14, pp. 157, 161.
- Variance and covariance: pp. 88–92; Exercises 5.12–5.13, pp. 108–109.
- Transformations: Exercise 7.10, p. 151.

Page numbers are the printed book pages. Additional proofs and reading boundaries are in the instructor guide.